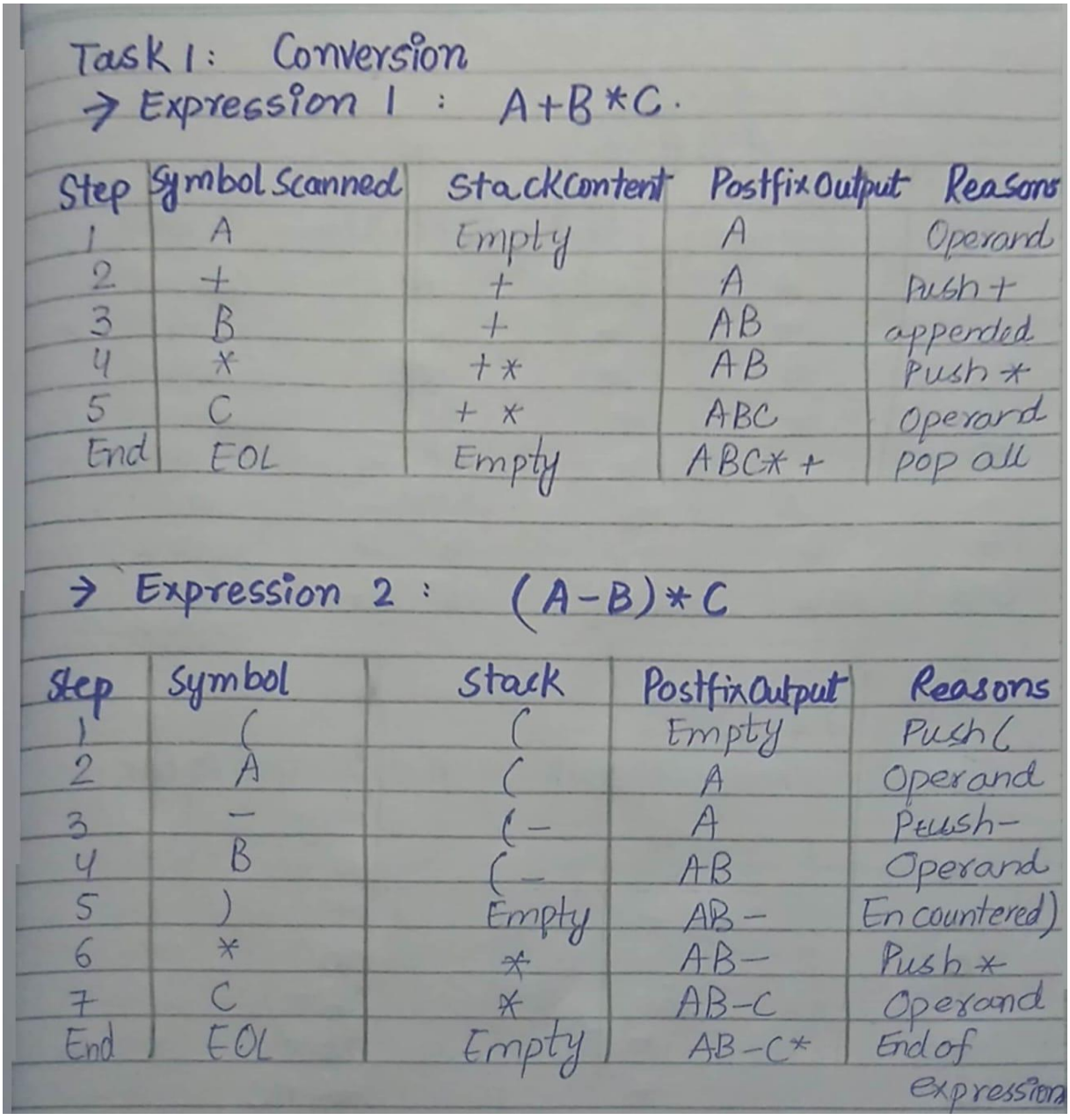
DSA Lab: 9

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Task 1:



Task 1: Conversion

→ Expression 1 : A+B*C.

Step	Symbol Scanned	Stack Content	Postfix Output	Reasons
1	A	Empty	A	Operand
2	+	+	A	Push +
3	B	+	AB	apperded
4	*	+*	AB	Push *
5	C	+*	ABC	Operand
End	EOL	Empty	ABC*+	pop all

→ Expression 2 : (A-B)*C

Step	Symbol	Stack	Postfix Output	Reasons
1	(	(	Empty	Push (
2	A	(	A	Operand
3	-	(-	A	Push -
4	B	(-	AB	Operand
5	)	Empty	AB-	Encountered)
6	*	*	AB-	Push *
7	C	*	AB-C	Operand
End	EOL	Empty	AB-C*	End of expression

→ Expression 3 :

A*B+C/D

Step	Symbol	Stack	Postfix Output	Reasons
1	A	Empty	A	Operand
2	*	*	A	Push *
3	B	*	AB	Operand
4	+	+	AB*	Scanned +
5	C	+	AB*C	Operand
6	/	+/	AB*C	push /
7	D	+/	AB*CD	Operand
End	EOL	Empty	AB*CD/+	End of expression.

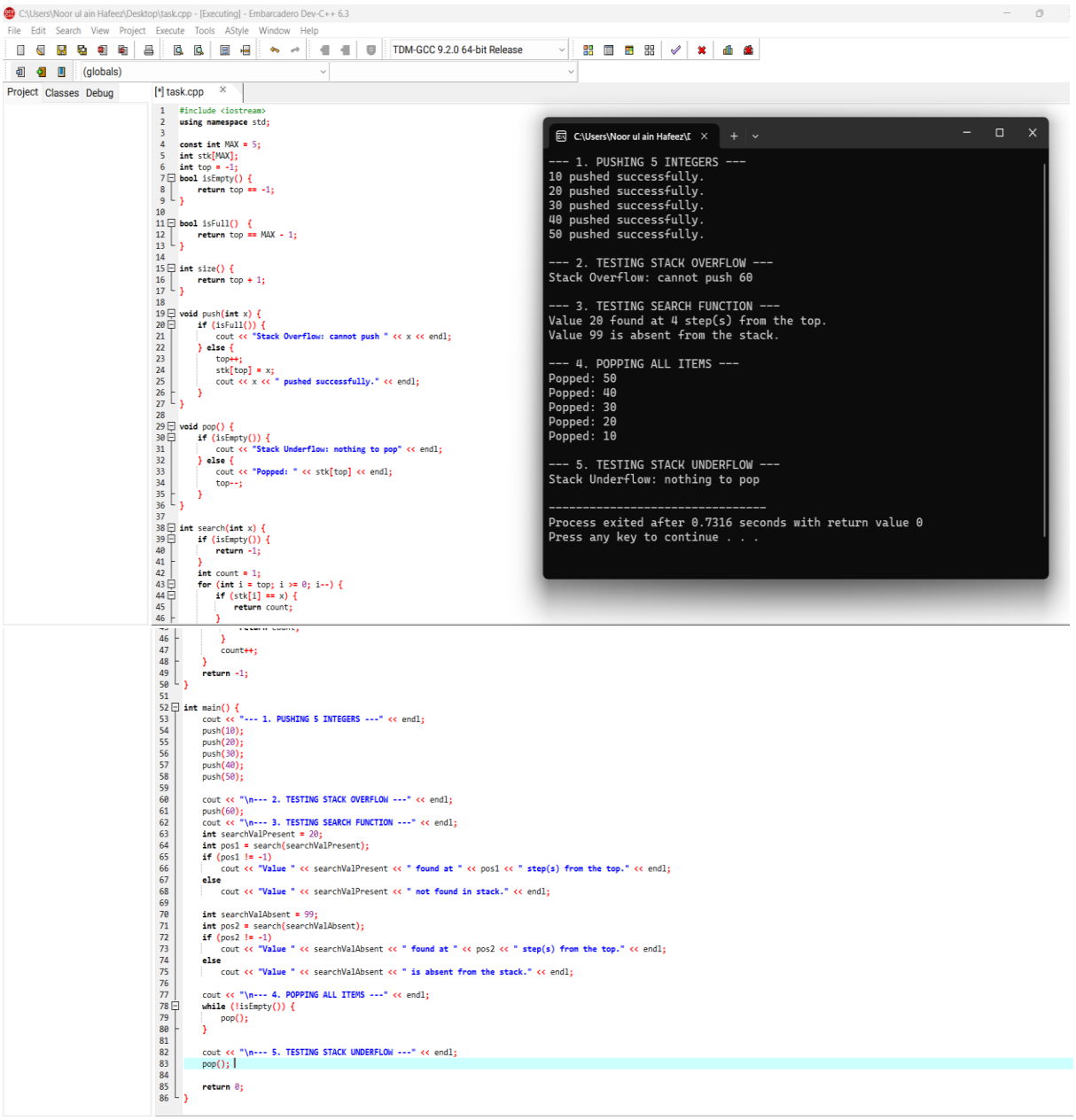
Task 2:

Task 2 :

(A+B)*C

Scanned	Contents of stack	Postfix Output so Far
(	(	Empty
A	(	A
+	(+	A
B	(+	AB
)	Empty	AB+
*	*	AB+
C	*	AB+C
End of Input	Empty	AB+C*

Task 3:




```cpp
#include <iostream>
using namespace std;

const int MAX = 5;
int stk[MAX];
int top = -1;
bool isEmpty() {
    return top == -1;
}

bool isFull()  {
    return top == MAX - 1;
}

int size() {
    return top + 1;
}

void push(int x) {
    if (isFull()) {
        cout << "Stack Overflow: cannot push " << x << endl;
    } else {
        top++;
        stk[top] = x;
        cout << x << " pushed successfully." << endl;
    }
}

void pop() {
    if (isEmpty()) {
        cout << "Stack Underflow: nothing to pop" << endl;
    } else {
        cout << "Popped: " << stk[top] << endl;
        top--;
    }
}

int search(int x) {
    if (isEmpty()) {
        return -1;
    }
    int count = 1;
    for (int i = top; i >= 0; i--) {
        if (stk[i] == x) {
            return count;
        }
        count++;
    }
    return -1;
}

int main() {
    cout << "--- 1. PUSHING 5 INTEGERS ---" << endl;
    push(10);
    push(20);
    push(30);
    push(40);
    push(50);

    cout << "\n--- 2. TESTING STACK OVERFLOW ---" << endl;
    push(60);
    cout << "\n--- 3. TESTING SEARCH FUNCTION ---" << endl;
    int searchValPresent = 20;
    int pos1 = search(searchValPresent);
    if (pos1 != -1)
        cout << "Value " << searchValPresent << " found at " << pos1 << " step(s) from the top." << endl;
    else
        cout << "Value " << searchValPresent << " not found in stack." << endl;

    int searchValAbsent = 99;
    int pos2 = search(searchValAbsent);
    if (pos2 != -1)
        cout << "Value " << searchValAbsent << " found at " << pos2 << " step(s) from the top." << endl;
    else
        cout << "Value " << searchValAbsent << " is absent from the stack." << endl;

    cout << "\n--- 4. POPPING ALL ITEMS ---" << endl;
    while (!isEmpty()) {
        pop();
    }

    cout << "\n--- 5. TESTING STACK UNDERFLOW ---" << endl;
    pop();

    return 0;
}
```